

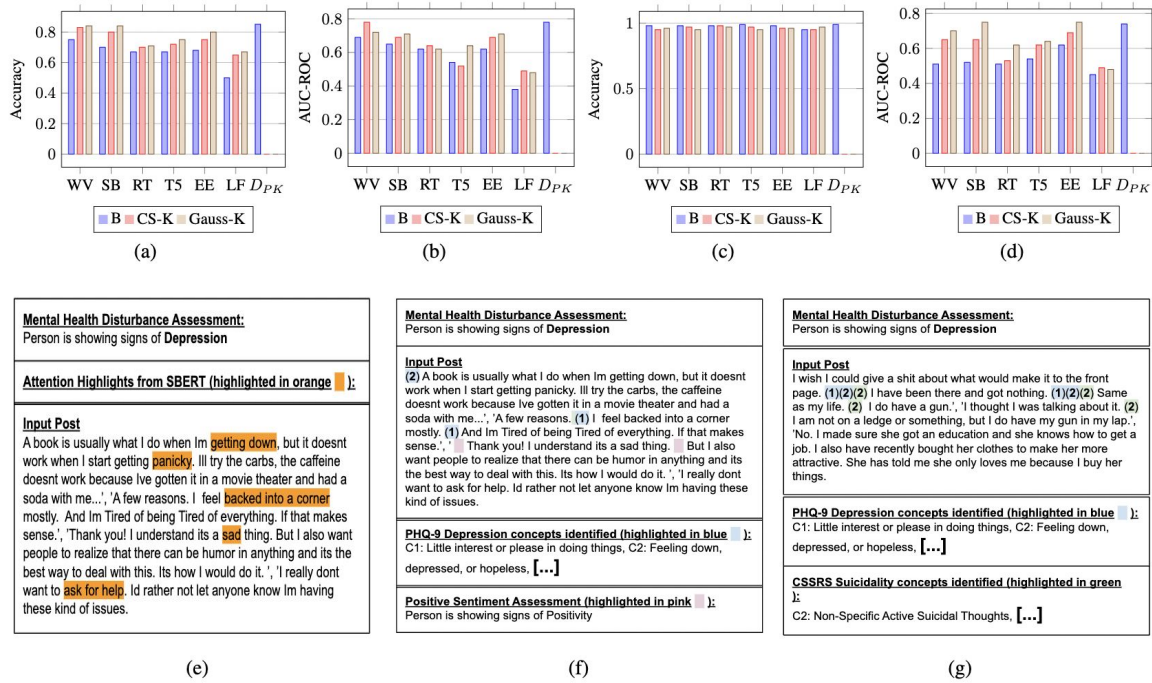


Figure 3: (a) and (b) present results for the CSSRS 2.0 dataset, while (c) and (d) show the results for the PRIMATE dataset. The mean accuracy/AUC-ROC of different language models (LMs) - Baseline fine-tuned model (B), PKiL performance with Cosine Similarity Kernel (CS-K), and PKiL performance with normalized Gaussian Kernel similarity (Gauss-K) - are displayed. The prompt-based model Text-Davinci-003_{PK} model (D_{PK}) doesn't utilize CS-K or Gauss-K, so no associated bar is shown. WV: Word2Vec, SB: SBERT, RT: RoBERTa, EE: ERNIE, LF: LongFormer. (d) The self-attention-based interpretability visualization for the SBERT baseline model indicates correct predictions and sensible highlights. However, the mapping of these highlights to clinician-friendly concepts used in practice is unclear. Baseline language models consistently struggle to capture negation. (e) The SB model trained with PKiL using the normalized Gaussian kernel provides clinicians with annotated explanations that are more familiar. Additionally, the PKiL parameters enable the analysis of fragments conveying positive sentiment. (f) Explanations from the Text-Davinci-003_{PK} model also demonstrate that leveraging process knowledge helps clinicians better understand the annotated explanations, as they are associated with familiar problem concepts.

form insights for refining existing process knowledge guidelines (e.g., adaptation to Reddit vocabulary) to facilitate remote monitoring and improved access to healthcare via social media platforms.

Reproducibility: We provide the trained model for SBERT with normalized Gaussian kernel similarity, the CSSRS 2.0 dataset, and the CSSRS process knowledge used in our experiments at the link in the footnote⁷. Additionally, we also provide a Python notebook for users to play with the Text-Davinci-003_{PK} model at the link in this footnote⁸.

Ethics Statement: We adhere to anonymity, data privacy, intended use, and practical implication of the AI-based mental health assessment systems. The clinical process knowledge does not contain personally identifiable information. The datasets covered in the survey are publicly available and

can be obtained from user-author agreement forms. Figures and examples are abstract and do not represent real-time data sources or any person.

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Possimus nam cumque, molestiae animi pariatul culpa aliquam, odit voluptatum eos quo sed quaerat molestiae quam consecetur harum, inventore tenetur porro iure dolorum commodi, nulla veritatis sequi tempore temporibus illum maxime illo. Molestiae perferendis aspernatur libero magni velit nostrum atque quae, repudiandae perferendis

⁷<https://anonymous.4open.science/r/MenatalHealthAnoynomous-8CC3/README.md>

⁸<https://anonymous.4open.science/r/MenatalHealthAnoynomous-8CC3/app.ipynb>

nobis animi est temporibus sapiente, velit laudantium quas
hic sit, praesentium minus laboriosam nemo officiis nostrum
perspiciatis, totam illo quisquam?